Machine Learning

16-08-26

In ML below math concepts are very useful
- Statistics ✓
- Linear Algebra
- calculus

Linear Algebra

- This branch of maths which deals with scalers, vectors, matrix, dot product, determinent, etc.

- In most of the ML and DL models, internal computation happen using matrixes

- Linear algebra is needed to understand these operations.

Scaler

- an entity having a value / magnitude

eg. $x = 5$ $\alpha = 0.95$
$y = -10$

scaler quantities
weight = 40 kg area = 1200 sq ft
time = 5.2 hrs

- scalers are used for measurement of single entity.

- scaler is also known as 0-dimensional array.

eg.

```
arr = np.array(10)
arr.shape = ( )
arr.ndim = 0
```

- scaler quantities don't have direction, they only have magnitude

Vector

- collection of multiple scalers
- collection of multiple 0-D arrays

vector is called 1-dimensional array.

eg.

$V_1 = [10, 20, 5]$

$V_2 = [10, 5]$

$V_3 = [-10, 0, 2]$

- vector is 1-D array and it exists in N-dimensional space

$V_1 = [1, 2]$ $V_2 = [2, 4]$ $V_3 = [-5, 2]$ $V_4 = [3, 7]$

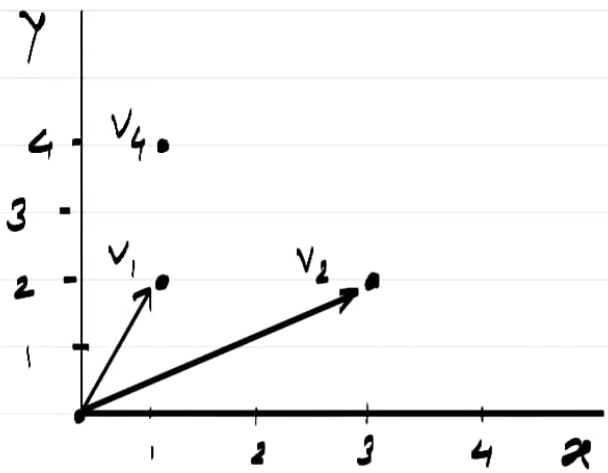
vector in 2 dimensional space

$V_1 : [4, 3, 5, 7]$ $V_2 = [-1, 0, 0, 1]$

vectors in 4 dimensional space

V_1 and V_2 both are 1-D arrays

- A vector represents a point in any given N-dimensional space.

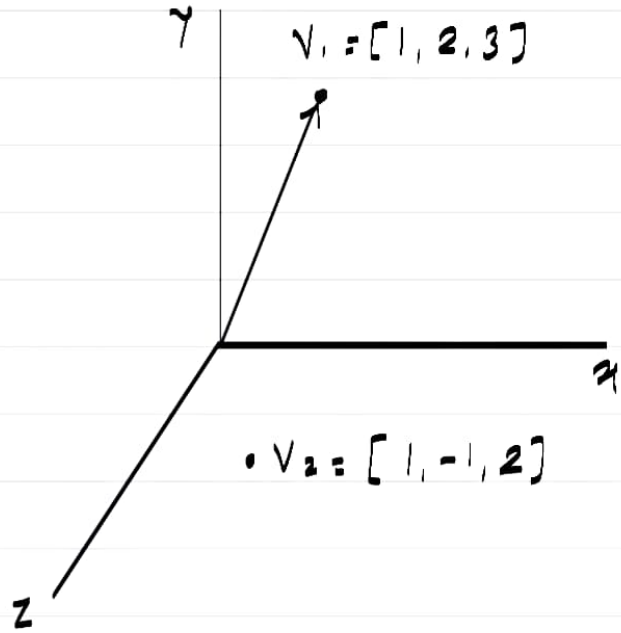


$V_1 : [1, 2]$

$V_2 = [3, 2]$

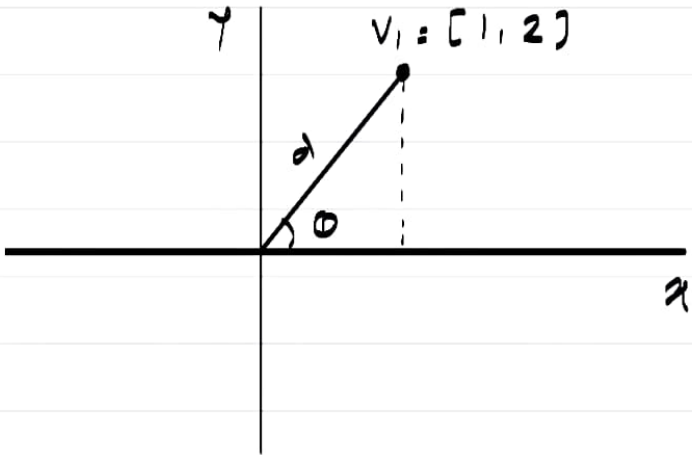
$V_4 : [1, 4]$

2D space



3D space

- vector has magnitude and direction

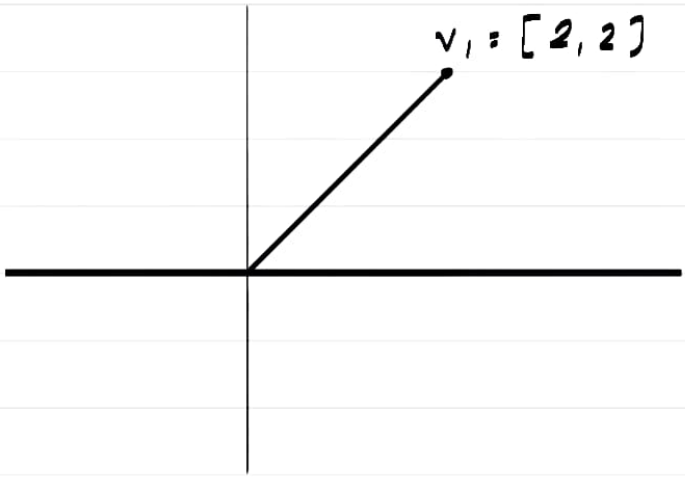


euclidean magnitude is distance betn origin and vector

$$d = \sqrt{1^2 + 2^2} = \sqrt{5}$$

direction is angle betn x axis and vector

$$\tan\theta = \frac{2}{1} = 2 \qquad \theta = \tan^{-1}(2)$$



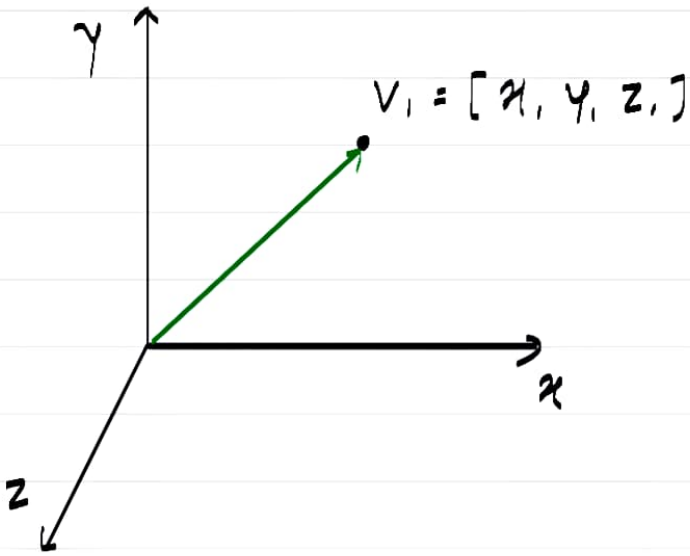
magnitude $= |v_1| = \sqrt{2^2 + 2^2}$

$|v_1| = \sqrt{8}$

direction

$\theta = \tan^{-1}\left(\frac{2}{2}\right) = \tan^{-1}(1)$

$\theta = 45°$



$|v_1| = \sqrt{x_1^2 + y_1^2 + z_1^2}$

for any vector in N-dimensional space

$V = [a_1 \quad a_2 \quad a_3 \ldots a_n]$

$|V| = \sqrt{a_1^2 + a_2^2 + a_3^2 \ldots a_n^2}$

eg.

house area	rooms	price
100	1	10
120	1	11
150	2	16

$V_1 = [100 \quad 1 \quad 10]$

$V_2 = [120 \quad 1 \quad 11]$

$V_3 = [150 \quad 2 \quad 16]$

V_1 V_2 V_3 are vectors in 3-D space

↓ individual datapoints

→ 3 columns in dataset

Matrix

- matrix is collection of multiple vectors
- matrix is collection of multiple 1-D arrays

matrix is a 2-D array

$V_1 = [1 \quad 2]$
$V_2 = [3 \quad 4]$
$V_3 = [-1 \quad 0]$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ -1 & 0 \end{bmatrix}_{3\times 2}$$

$$B = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 4 & 0 \end{bmatrix}_{2\times 3}$$

every row and every column in a matrix represents a vector

eg.

house area	rooms	price
100	1	20
160	2	25
200	3	45

$$\text{data matrix} = \begin{bmatrix} 100 & 1 & 20 \\ 160 & 2 & 25 \\ 200 & 3 & 45 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

scaler	0-D array	magnitude, no direction
vector	1-D array	magnitude, direction
matrix	2-D array	determinant, shape
tensor	3-D array → collection of multiple matrix	

Scaler, vectors, Matrix operations

scaler - scaler operations

$a = 5$ $\quad$ $b = 4$

$a + b = 9$ $\quad$ $a \times b = 20$

$a - b = 1$ $\quad$ $a \div b = 1.25$

scaler - vector operations

$a = 2$ $\quad$ $V_1 = [4 \quad 3 \quad 1]$

$V_1 \times a = [8 \quad 6 \quad 2]$

$V_1 \div a = [2 \quad 1.5 \quad 0.5]$

vector - vector operations

$\vec{V_1} = [1 \quad 2]$ $\quad$ $\vec{V_2} = [3 \quad 4]$

$V_1 + V_2 = [4 \quad 6]$
$V_1 - V_2 = [-2 \quad -2]$ } elementwise operations

for these operations, both vectors should have same size i.e. same number of elements

$V_1 = [1 \quad 2 \quad 3]$ $\quad$ $V_2 = [4 \quad 5]$

$V_1 + V_2 \rightarrow$ not possible

$V_1 - V_2 \rightarrow$ not possible

dot product of vectors

$V_1 = [1 \quad 2]$ $\quad$ $V_2 = [3 \quad 4]$

$V_1 . V_2 = 1 \times 3 + 2 \times 4 = 3 + 8 = 11$

$V_1 = [-1 \quad 2] \qquad V_2 = [3 \quad 4]$

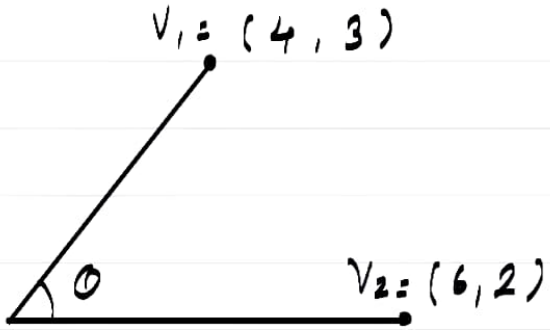
$$\underbrace{V_1 \cdot V_2}_{\text{vectors}} = -1 \times 3 + 2 \times 4 = -3 + 8 = \underset{\text{scaler}}{5}$$

result of dot product of vectors is a scaler

$V_1 = [a_1 \quad a_2 \quad a_3 \ldots a_n] \qquad V_2 = [b_1 \quad b_2 \quad b_3 \ldots b_n]$

$$V_1 \cdot V_2 = \sum_{i=1}^{n} a_i b_i$$

dot product is very useful in finding relationship in two vectors



$$V_1 \cdot V_2 = |V_1| \, |V_2| \cos(\theta)$$

$$\cos(\theta) = \frac{V_1 \cdot V_2}{|V_1| \, |V_2|}$$

$$\cos(\theta) = \frac{24 + 6}{\sqrt{25} \times \sqrt{40}} = \frac{30}{5 \times \sqrt{40}} = \frac{6}{\sqrt{40}}$$

if angle θ is small $\rightarrow$ V_1 and V_2 are similar or closer to each other

$$\cos(\theta) = \frac{V_1 \cdot V_2}{|V_1| \; |V_2|}$$

$\downarrow$

cosine similarity

$$-1 \leq \cos(\theta) \leq 1$$

higher $\cos(\theta)$ means similarity between vectors is very high

RAG pipelines are based on this cosine similarity

docs

$V_1 \leftarrow d_1$

$V_2 \leftarrow d_2$

$V_3 \leftarrow d_3$

$V_4 \leftarrow d_4$

$V_5 \leftarrow d_5$

query $\rightarrow$ chatbot

$\rightarrow q_1$

$$\left.\begin{aligned} C_1 &= \frac{q_1 \cdot V_1}{|q_1| \, |V_1|} \\ C_2 &= \frac{q_1 \cdot V_2}{|q_1| \, |V_2|} \\ &\vdots \\ C_5 &= \frac{q_1 \cdot V_5}{|q_1| \, |V_5|} \end{aligned}\right\}$$

doc with highest cosine similarity will be used to answer the query